Choose Informative Questions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Choose Informative Questions before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Accuracy as a Fraction

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Features and Distractors

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Choose Informative Questions trace and record the first step you can explain without help.

Use W2: Verify the candidate scores for **Fitting** a decision tree. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Choose Informative Questions explanation against the supplied facts before discussing it.

Use the worked case for **Fitting** a decision tree. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Ten cards have six pet subjects and four space subjects; all are green. Compare asking “green?” with asking “pet?” to identify a card. Give the largest remaining group and one limitation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

Connect the Choose Informative Questions target to a trace and a justified conclusion.

Six cards contain three animal and three space books; all have blue covers. Compare asking 'blue cover?' with 'animal subject?'. Is the better question guaranteed to work on inaccurately tagged books?

WORKBOOK · 6 OF 21

Solve the given case

Solve the supplied Choose Informative Questions case and show the deciding evidence.

Eight blue cards contain four circles and four triangles; target is shape.

> Which question is more useful: blue or circle?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Cards are two cats and two dogs, all red. Which question helps separate species: is it red or does it bark? State your assumption.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Choose Informative Questions adds to Transparent Book Recommender and keep the evidence.

Ask first about preferences that remove the largest number of unsuitable books.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Six cards contain three animal and three space books; all have blue covers. Compare asking 'blue cover?' with 'animal subject?'. Is the better question guaranteed to work on inaccurately tagged books?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Name the parts and decisions

Name the parts and decisions. Apply the supplied Choose Informative Questions facts and keep your reasoning.

Identify the **root**, its two **branch**es, **leaf prediction**s and **depth** of the worked fitted tree. Which parts were selected using **training** labels?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Verify the candidate scores

> Verify the candidate scores. Apply the supplied Choose Informative Questions facts and keep your reasoning.

For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Separate fitting from prediction

> Separate fitting from prediction. Apply the supplied Choose Informative Questions facts and keep your reasoning.

A learner sees a blocked validation case with near=0 and changes that **leaf** to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Fit a new tree independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Fit a new tree independently. Apply the supplied Choose Informative Questions facts and keep your reasoning.

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Audit a tie and a claim

> Audit a tie and a claim. Apply the supplied Choose Informative Questions facts and keep your reasoning.

If near and tall make the same **training** error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Choose Informative Questions

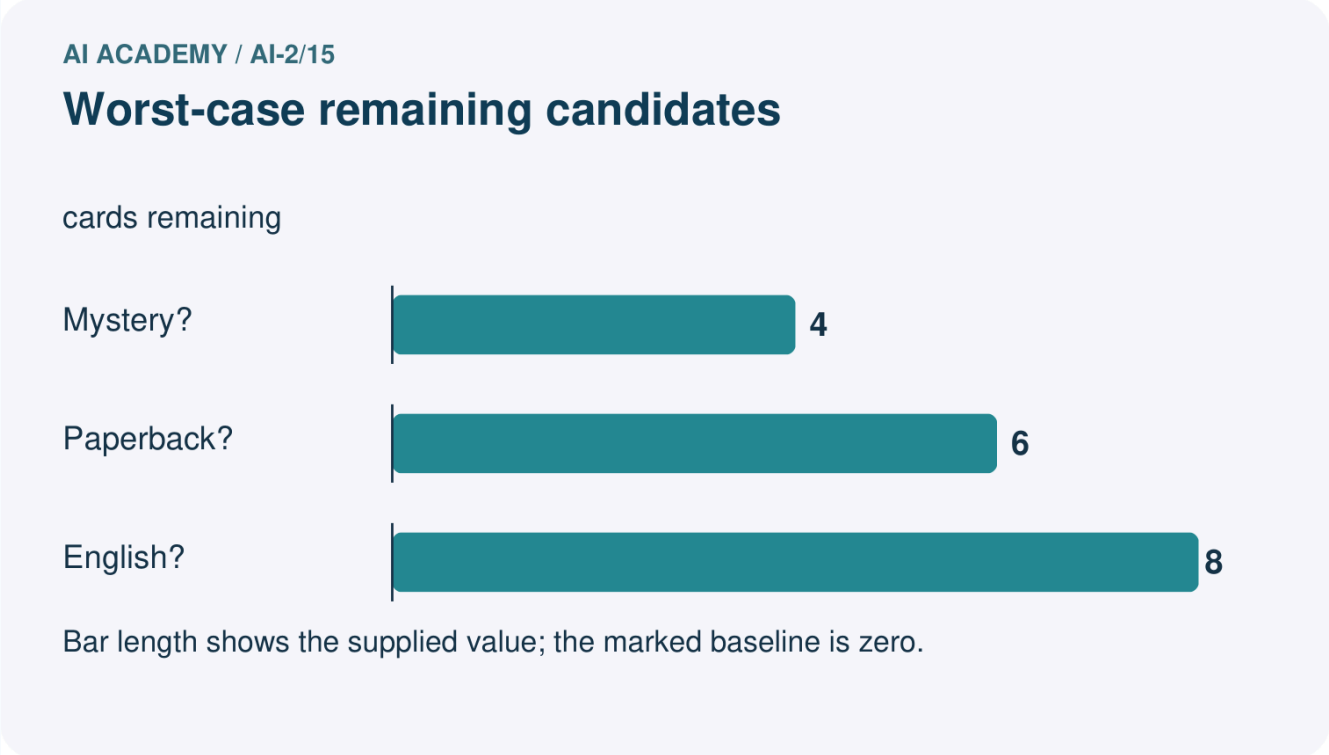
> A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the

> target. Explain the labelled step before checking the changed condition.

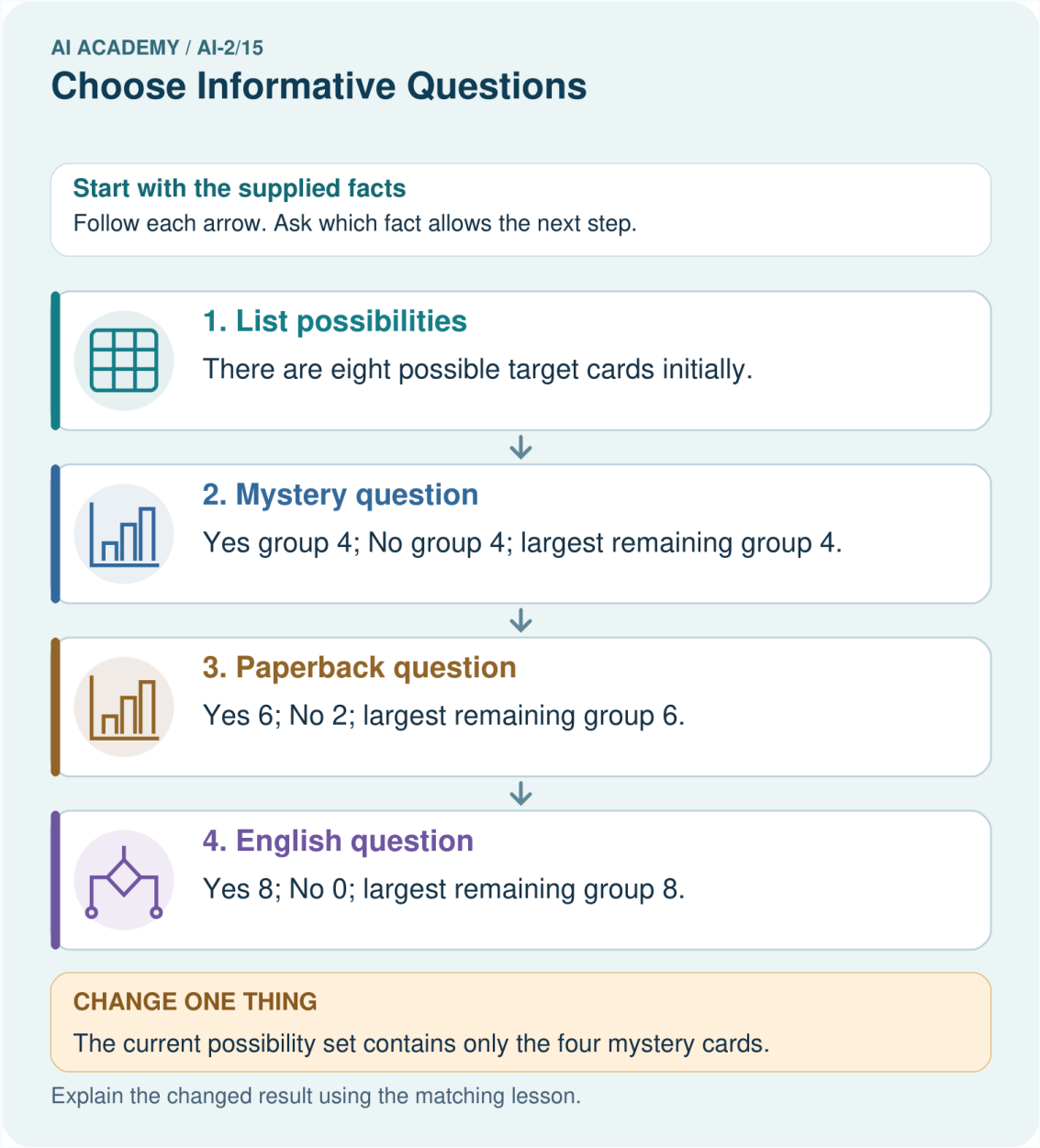
A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

Cards: four mystery, four science; six paperback, two hardback; all English.

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose the best original question and explain why its more balanced **split** matters here.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Yes 8; No 0; largest remaining group 8.

1. List possibilities

2. Mystery question

3. Paperback question

4. English question

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: English question Yes 8; No 0; largest remaining group 8.

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The current possibility set contains only the four mystery cards. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Asking mystery now leaves all four; it no longer narrows the set.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Asking mystery now leaves all four; it no longer narrows the set. Question value depends on the remaining candidates, not just on its wording.

Repair a missing building block

Review Decision Trees (ai-2/14). Goal: Fit a **root** question and **leaf prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

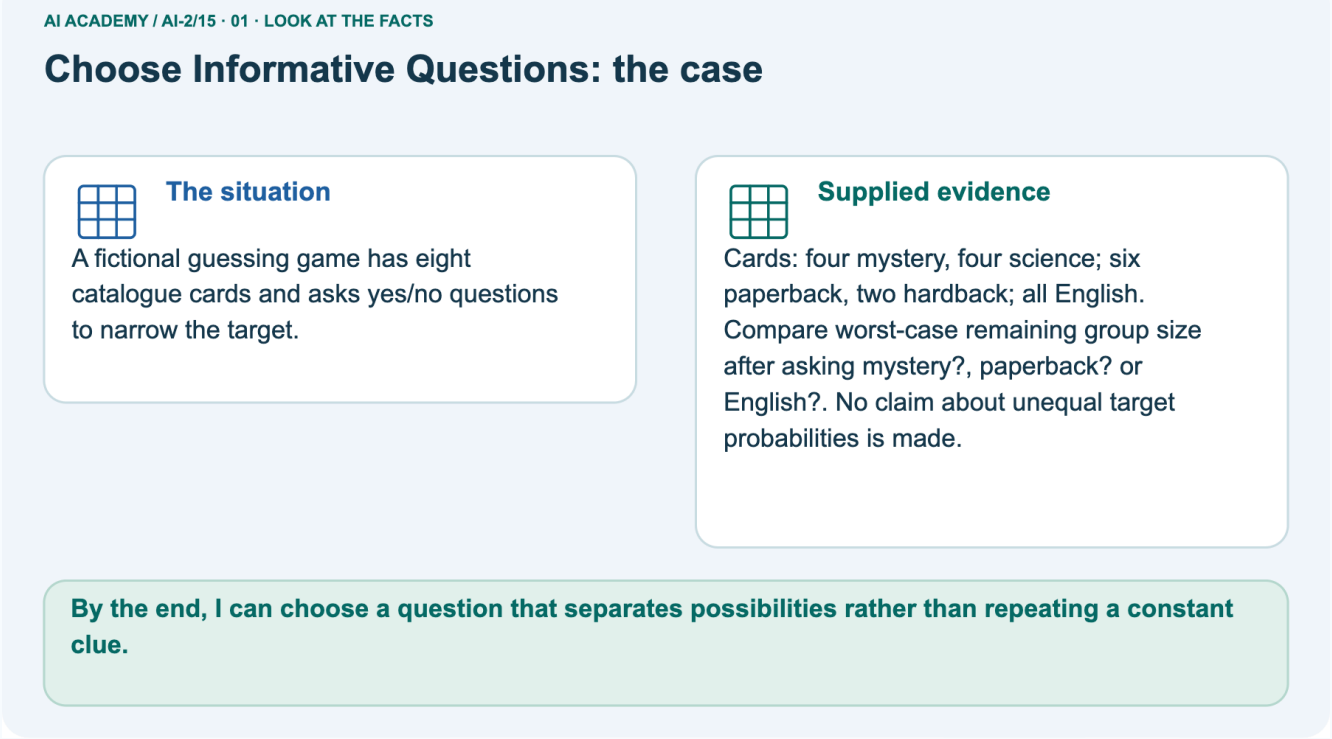
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Choose Informative Questions picture from its labels, then record a relevant limit.

01. Choose Informative Questions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

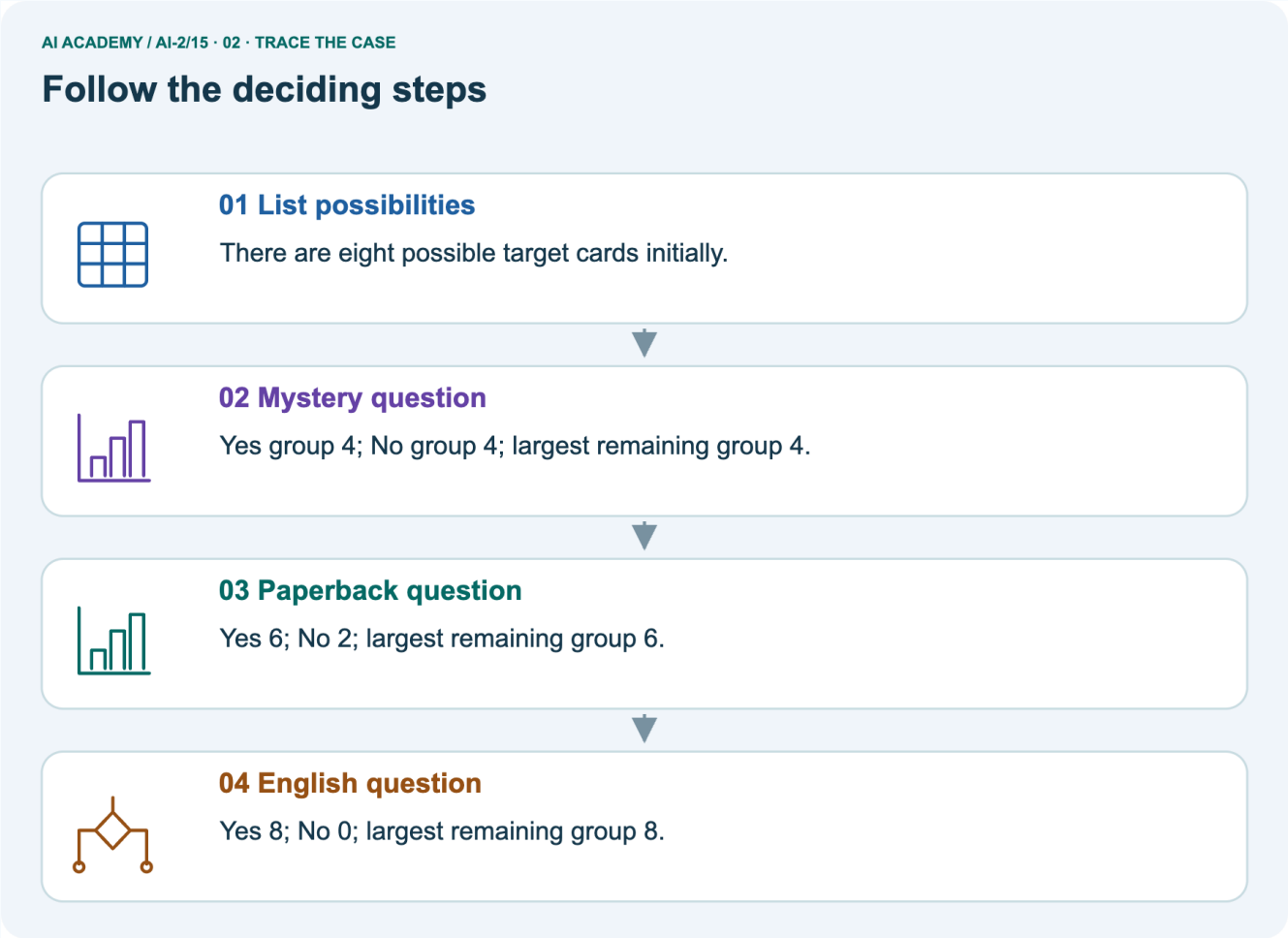
A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target. Supplied facts: Cards: four mystery, four science; six paperback, two hardback; all English.

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



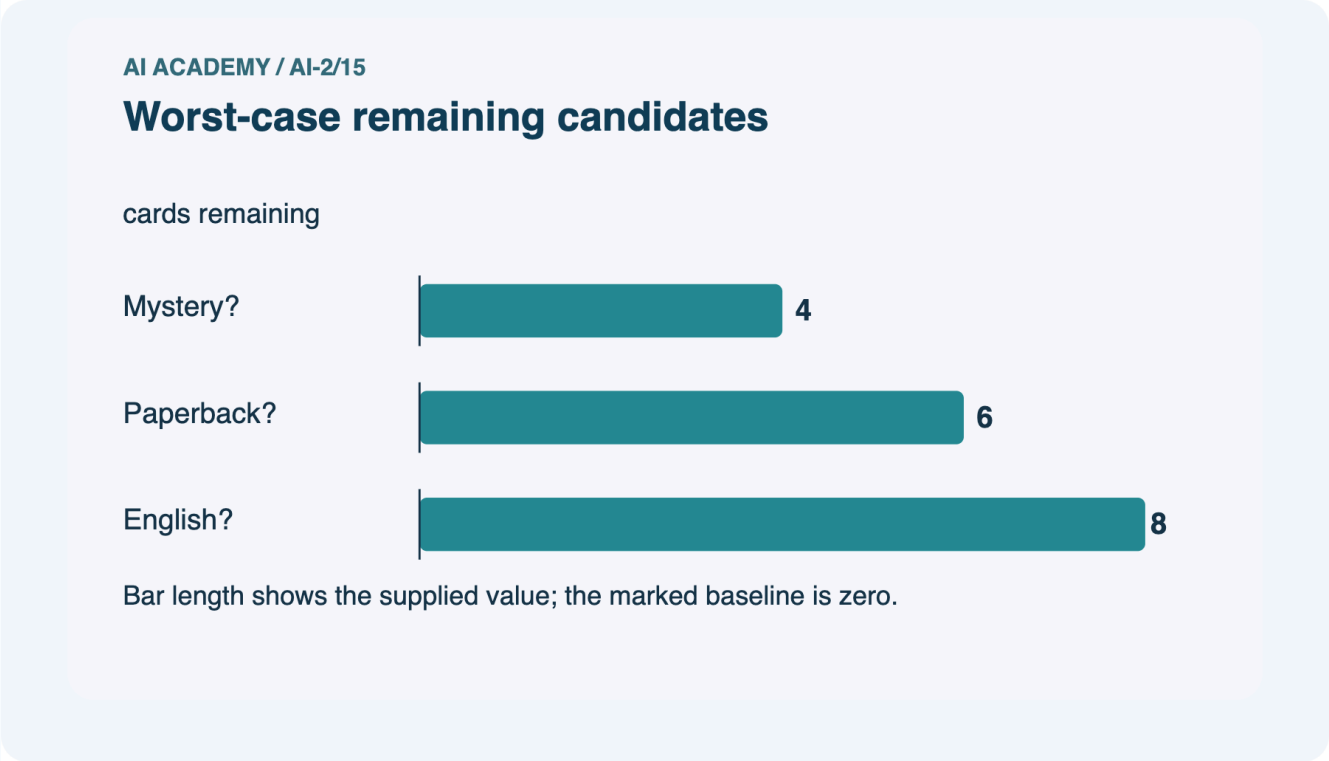
Follow the labelled picture with the matching explanation. Open the vector diagram

List possibilities: There are eight possible target cards initially. → Mystery question: Yes group 4; No group 4; largest remaining group 4. → Paperback question: Yes 6; No 2; largest remaining group 6. → English question: Yes 8; No 0; largest remaining group 8.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Worst-case remaining candidates



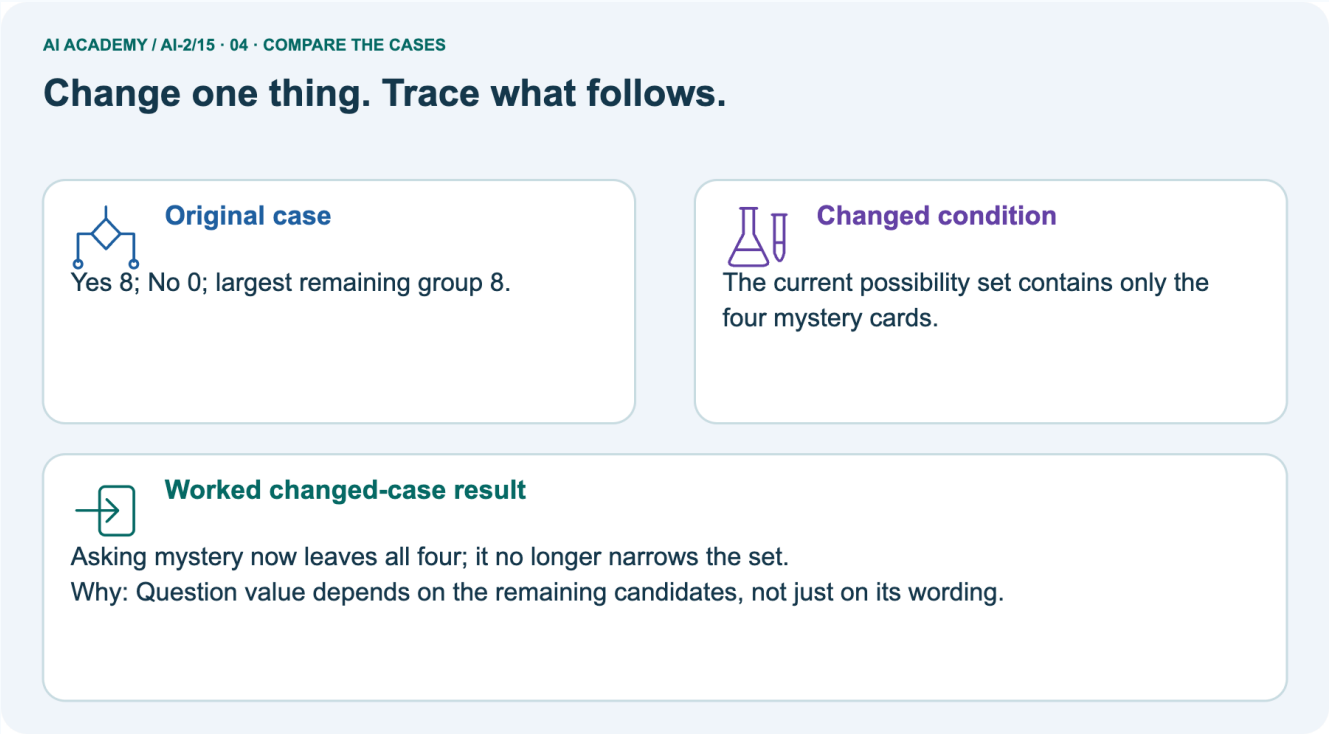
Follow the labelled picture with the matching explanation. Open the vector diagram

Worst-case remaining candidates. Units: cards remaining. Mystery?: 4. Paperback?: 6. English?: 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Yes 8; No 0; largest remaining group 8. Changed condition: The current possibility set contains only the four mystery cards. Expected result: Asking mystery now leaves all four; it no longer narrows the set.

Reason: Question value depends on the remaining candidates, not just on its wording.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit

AI ACADEMY / AI-2/15 · 05 · EXPLAIN WITH CARE

Connect the explanation to its limit

What the case explains

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set. This is an identification-game criterion; a classifier fitting labels may instead compare prediction errors or another declared training measure.

What it does not establish

Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

Follow the labelled picture with the matching explanation. Open the vector diagram

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

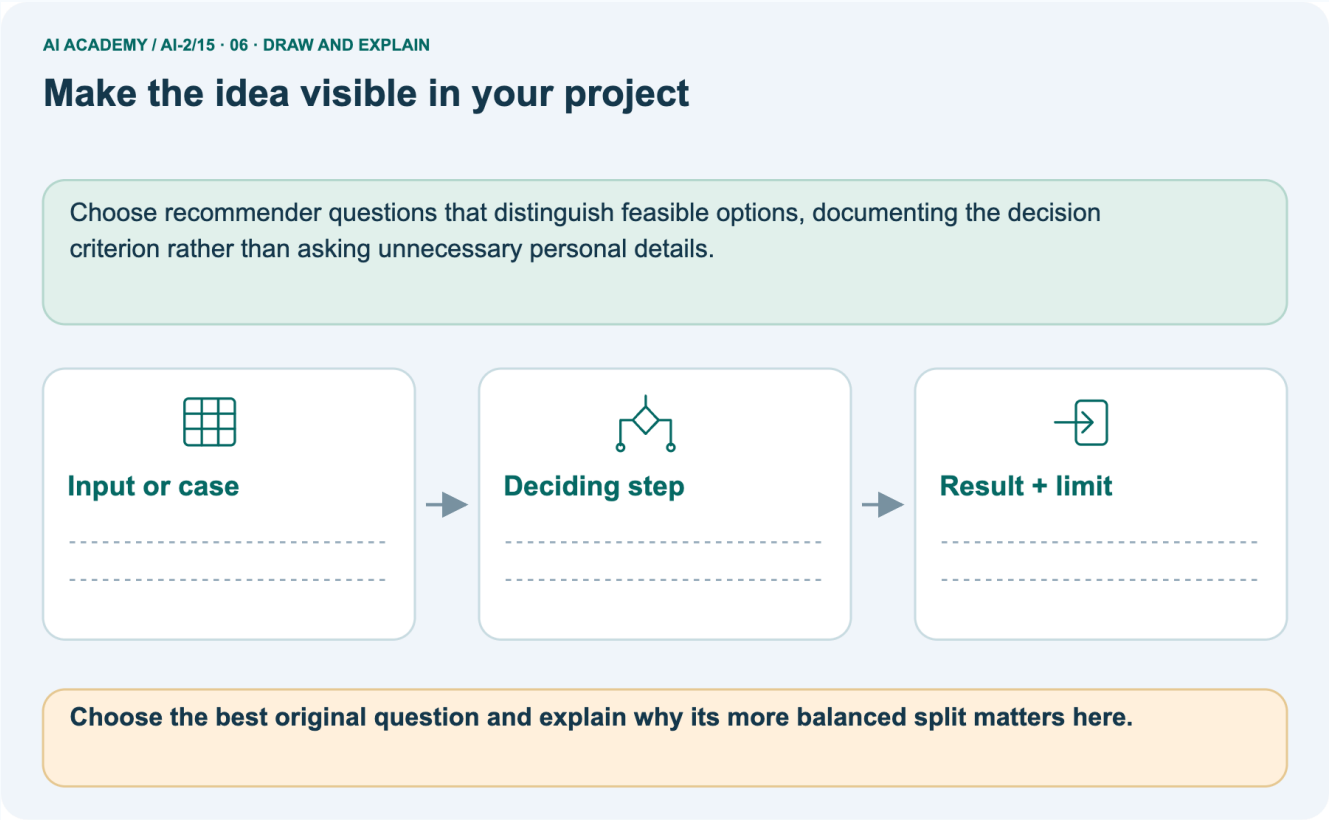
This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

Limit: Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details. Draw your **input**, deciding step, result and limit.

Supported question: Choose the best original question and explain why its more balanced **split** matters here.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.